

LOGIC BOMB

DISCRETE MATH EDITION

BOMB DEFUSAL MANUAL

Version 2.1

! How to Play

DEFUSER

- Sees the bomb on the computer.
- Describes what is on it.
- Presses buttons and cuts wires.
- **Never** looks at this manual.

EXPERT

- Reads this manual.
- Does the math.
- Tells the Defuser what to do.
- **Never** looks at the screen.

THE RULES

1. **No peeking.** The Defuser and the Expert can't see each other's side. Talking is the only way to win.
2. **Beat the clock.** Disarm every module before the timer hits 0:00.
3. **Watch your strikes.** Every mistake is a strike and makes the timer faster. Too many strikes and the bomb explodes.
4. **Any order.** Do the modules in any order. A module is done when its light turns **green**.
5. **First true rule wins.** When a module lists rules, check them top to bottom and use only the first one that is true.

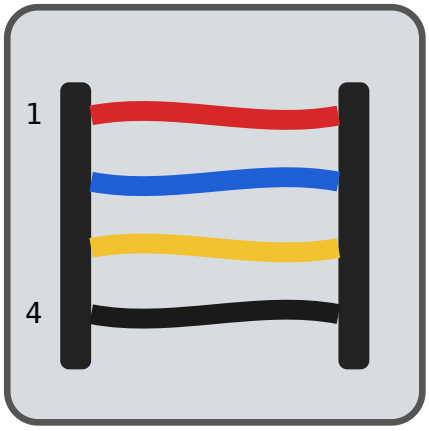
READING THE CASING (TURN THE BOMB TO SEE ITS SIDES)

ON THE CASING	WRITE DOWN
Serial number	**D** = set of its digits · **V** = set of its vowels
Batteries	**b** = number of batteries
Indicators	Is any indicator **lit**?

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1 Wires



1. Count the wires (3–6).
2. Wire 1 is the **top** wire. “Last” = lowest.
3. Cut **one** wire.

R(w) red · **B(w)** blue · **Y(w)** yellow · **K(w)** black · **W(w)** white

3 WIRES

1	¬∃w R(w)	**cut wire 2**
2	∀w (R(w) ∨ B(w))	**cut last wire**
3	∃w W(w)	**cut last white**
4	otherwise	**cut wire 1**

4 WIRES

1	∀w ¬Y(w)	**cut wire 1**
2	∃w K(w)	**cut last black**
3	∀w (R(w) ∨ Y(w))	**cut wire 3**
4	otherwise	**cut wire 4**

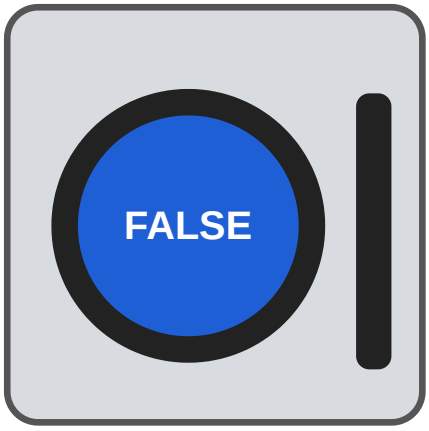
5 WIRES

1	∃w K(w) ∧ ∃w W(w)	**cut wire 4**
2	∀w ¬R(w)	**cut wire 1**
3	∃d ∈ D (d is odd)	**cut last wire**
4	otherwise	**cut wire 2**

6 WIRES

1	∀w ¬Y(w)	**cut wire 3**
2	∃w W(w)	**cut last white**
3	∃w ∃v (w ≠ v ∧ R(w) ∧ R(v))	**cut last red**
4	otherwise	**cut wire 6**

2 The Button



1. Find the first TRUE rule.
2. **TAP** = press and let go.
3. **HOLD** = keep holding; a strip lights up. Release using the table below.

p	button is red
q	label says TRUE
r	2 or more batteries
s	an indicator is lit

RULES

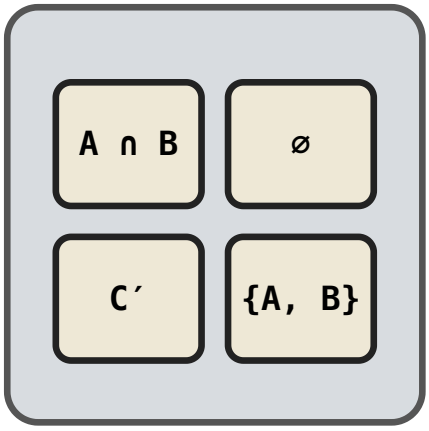
1	**p ∧ q**	**TAP**
2	**¬p ∧ ¬r**	**HOLD**
3	**q ∨ s**	**TAP**
4	otherwise	**HOLD**

RELEASE WHEN THE TIMER SHOWS…

STRIP	THE DIGIT IN
Blue	**{1,2,3,4} ∩ {4,5,6}**
White	**{1,3,5} − {3,5}**
Yellow	**{5,6,7} ∩ {1,3,5}**
Red	**{x ∈ ℤ | 1 < x < 3}**

The digit can be anywhere on the timer.

3 Set Keypad



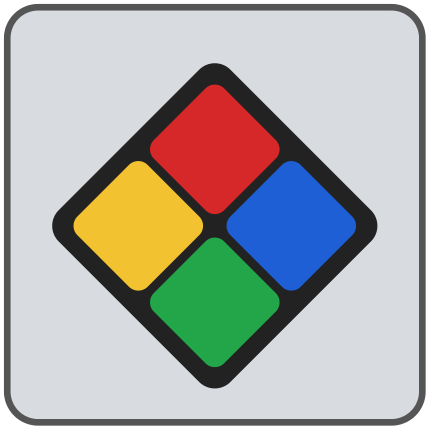
1. Each key shows a set.
2. Find each set’s size **|S|**.
3. Press keys from **smallest** to **largest**.

A = {1, 2, 3, 4} · **B = {3, 4, 5}** · **C = {2, 4, 6, 8}**
Universe **𝕌 = {1, …, 8}** (for complements like **A′**)

WORKSHEET

KEY SHOWS	ELEMENTS	SIZE	ORDER

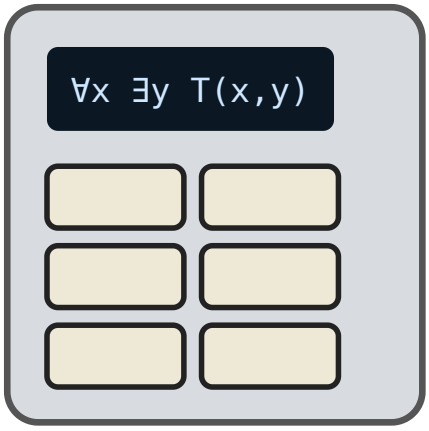
4 Simon Maps



1. Colors flash in a sequence that grows each stage.
2. For each flash **x**, press **f(x)**, in order.
3. Repeat the whole sequence every stage.

SERIAL	FUNCTION
has a vowel **(V ≠ ∅)**	**f = {(Red, Blue), (Blue, Red), (Green, Yellow), (Yellow, Green)}**
no vowel **(V = ∅)**	**f = {(Red, Yellow), (Blue, Green), (Green, Blue), (Yellow, Red)}**

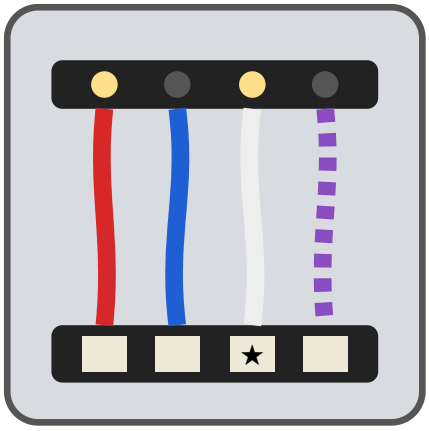
5 Who’s on ∀irst



1. **`T(x, y)`** = “x is taller than y”. Domain: the class.
2. Translate the screen into English.
3. Press that sentence’s code word. 3 stages.

SENTENCE	CODE WORD
Someone is taller than everyone.	**COMET**
Everyone has someone taller than them.	**EMBER**
No one is taller than anyone.	**GLACIER**
Everyone is taller than someone.	**BISHOP**
Someone is taller than someone.	**DYNAMO**
Someone has everyone taller than them.	**FALCON**
Everyone is taller than everyone.	**ATLAS**
Someone is taller than no one.	**HARBOR**

6 Venn Wires



1. Wires are numbered **left to right**.
2. For each wire: color, LED on/off, ★ or blank.
3. Cut every wire in K. Order doesn’t matter.

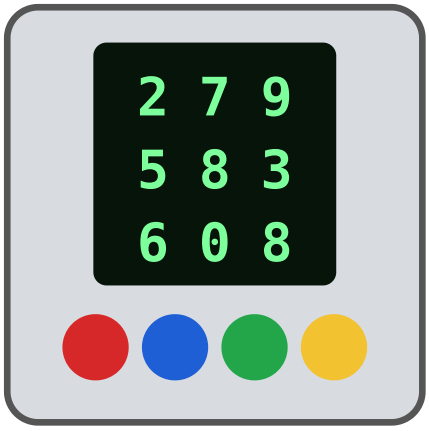
R	red or striped
B	blue or striped
L	LED on
S	★ below

$$K = (R \cap L) \cup (S - B)$$

WORKSHEET

#	R	B	L	S	CUT?
1					
2					
3					
4					
5					
6					

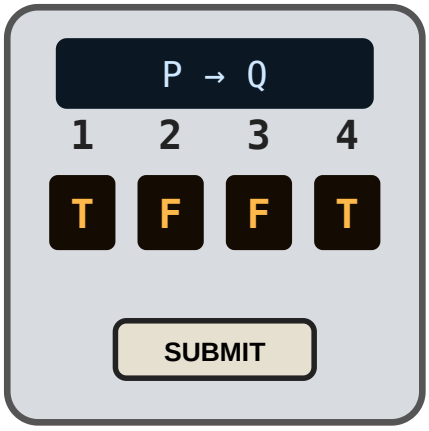
7 Quantifier Grid



1. Rows **r**: top to bottom. Columns **c**: left to right.
2. **x** = the digit at row r, column c.
3. Press the color of the first TRUE rule.

1	`∃r ∀c (x is even)`	**RED**
2	`∀r ∃c (x = 0)`	**BLUE**
3	`∃c ∀r (x > 5)`	**GREEN**
4	otherwise	**YELLOW**

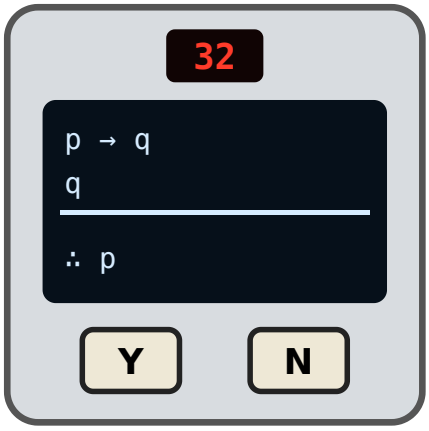
8 Truth Table



1. Set each window to the formula’s value for its row.
2. Arrows flip T ↔ F.
3. Press **SUBMIT**.

WINDOW	P	Q	FORMULA
1	T	T	
2	T	F	
3	F	T	
4	F	F	

N Argument Check (needy)



1. Wakes up now and then with a 40-second countdown.
2. **Y** = valid · **N** = invalid.
3. Never disarms. Keep answering it.

VALID → Y
p → q, p ∴ q
p → q, ¬q ∴ ¬p
p → q, q → r ∴ p → r
p ∨ q, ¬p ∴ q
p ∴ p ∨ q
p ∧ q ∴ p
∀x P(x) ∴ P(c)
P(c) ∴ ∃x P(x)
∀x (P(x) → Q(x)), P(c) ∴ Q(c)
¬∀x P(x) ∴ ∃x ¬P(x)

INVALID → N
p → q, q ∴ p
p → q, ¬p ∴ ¬q
p ∨ q, p ∴ ¬q
∀x (P(x) → Q(x)), Q(c) ∴ P(c)
∃x P(x) ∴ ∀x P(x)
¬∀x P(x) ∴ ∀x ¬P(x)

✓ Answer Key

Casing for every example: serial **KA4T74** (**D = {4, 7}**, **V = {A}**), 2 batteries, NEG lit.

MODULE	BOMB SHOWS	WORK	ANSWER
Wires	red, yellow, black, blue	Rule 1 F (yellow). Rule 2: a black exists.	**cut wire 3**
Wires	red, yellow, blue, black, blue	Rule 1 F (no white). Rule 2 F. Rule 3: 7 is odd.	**cut wire 5**
Button	blue, “NOT”	p F, q F, r T, s T. Rule 2: T ∧ F = F. Rule 3: F ∨ T = T.	**TAP**
Button	held, yellow strip	**{5,6,7} ∩ {1,3,5} = {5}**	**release on 5**
Keypad	A ∪ B, ∅, C − B, B ∩ C	sizes 5, 0, 3, 1	**∅, B∩C, C−B, A∪B**
Simon	red; then red, green	V ≠ ∅ → first function	**blue; blue, yellow**
∀irst	**∀x ∃y T(x,y)**	Everyone is taller than someone.	**BISHOP**
∀irst	**∃y ∀x T(x,y)**	Someone has everyone taller than them.	**FALCON**
Venn	1 red/LED · 2 blue/LED/★ · 3 white/★ · 4 striped/LED · 5 red/★ · 6 blue	**R∩L = {1,4}, S−B = {3,5}**	**cut 1, 3, 4, 5**
Grid	7 1 3 / 8 2 1 / 9 5 1	Column 1 = 7, 8, 9: all > 5.	**GREEN**
Truth	**¬P → Q**	TT: T · TF: T · FT: T · FF: F	**T T T F**
Argument	**∀x (P(x) → Q(x)), Q(c) ∴ P(c)**	Affirming the conclusion.	**N**